Mathematical Model

Transfer Function

The linearized aircraft longitudinal pitch transfer function relating elevator deflection δe(s) to pitch angle θ(s) is:

$$\frac{\boldsymbol{\theta(s)}}{\boldsymbol{\delta_e(s)}} = \frac{\boldsymbol{K(\tau s+1)}}{\boldsymbol{s^3 + a_2 s^2 + a_1 s + a_0}}$$

With the following parameter values:

Table 1 : Aircraft Pitch Transfer Function Parameters

Parameter	Symbol	Value	Physical Meaning
Gain	**K**	−1.282	Control effectiveness (negative = nose down for positive δe)
Zero time constant	**τ**	−1.0	RHP zero location at s = +1/|τ| = +1
Pitch damping coeff.	**a_2**	1.935	Aerodynamic pitch rate damping
Speed stability coeff.	**a_1**	0.987	Speed-pitch coupling derivative
Static stability coeff.	**a_0**	0.179	Static pitch restoring moment

Numerator Expansion:

$$K \times [\tau \quad 1] = -1.282 \times [-1 \quad 1] = [1.282 \quad -1.282]$$

Numerical Transfer Function:

$$\boldsymbol{G(s) = \frac{1.282s - 1.282}{s^3 + 1.935s^2 + 0.987s + 0.179}}$$